Aksa Fathima

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Project Associate at the BEES Lab, DESE, IISc, with hands-on research experience in nano-biosensors and point-of-care diagnostic systems. Trained in biosensor design and development, micro/nanofabrication, surface engineering, materials characterization and electrical impedance spectroscopy (EIS) based characterization. Research interests include, but are not limited to, BioMEMS and nano-biosensors, novel sensing platforms, nanoscale interfaces for biomolecule detection, and dynamic EIS approaches for single-molecule and ultra-low-concentration biosensing. Seeking a doctoral research position in advanced micro/nanofabrication technologies for translational biosensing platforms.

EDUCATION

- **Master of Science in Nanoscience and Technology** *2020 – 2022*

  JSS Academy of Higher Education and Research (JSSAHER), Deemed University, Mysore, India
  - CGPA: 8.3/10
  - Advisor: **Dr Asha Srinivasan**
  - Research Thesis: PVA Nanofibers for transdermal drug delivery of capsaicin and CBD for trigeminal neuralgia pain management.
- **Bachelor's in Health Care and Technology** *2016 – 2019*

  ST. Philomeana's Autonomous College, Affiliated to the University of Mysore, India
  - CGPA: 8.8/10 (secured third rank in the batch)
  - An interdisciplinary undergraduate program that combines cardiac science, medical biochemistry and microbiology (majors) along with a clinical internship in the cardiology department for about 2.5 years.

RESEARCH OUTCOMES: JOURNAL PUBLICATIONS

[1] Twinkle, H Madhumathi, **Fathima Aksa**, Vishnu Anil, Suresh Amritha, Das Manjula, Pandya Hardik, "**Impedance Profiling of Lymph Nodes: A Biomarker for Oral Cancer Metastasis"**, 2026 Elsevier Microchemical Journal, 227, p. 118842. https://doi.org/10.1016/j.microc.2026.118842

[2] **Fathima Aksa et al**. **"Smart use of nanomaterials as sensors for detection and monitoring of food spoilage"**, 2022 Application of nanotechnology in food science, processing, and packaging, 169-18. Springer International Publishing; https://doi.org/10.1007/978-3-030-98820-3_1

RESEARCH OUTCOMES: CONFERENCE PROCEEDINGS

[1] **Aksa Fathima**, Twinkle, Varun and Hardik J. Pandya, "**Towards Early Diagnosis of Congestive Heart Failure using multiplexed Immunosensor**." International Conference on Cardiology, Heart Disease and Cardiovascular Medicine (ICCHDCM- October 2025), Vienna, Italy. [Best Oral Presentation Award].

[2] Twinkle, Anil Vishnu, Madhumati H, **Aksa Fathima**, Vijay Pillai, Manjula Das, Amritha Suresh and Hardik J. Pandya, **"Electrical Impedance as a Signature for Metastatic Lymph Node Detection in Oral Cancer."** International Conference on Biomedical Science and Engineering (ICBSE-2025), Rome, Italy. [Oral Presentation]

[3] Sushma N.Moode, Hemanjali Mude, **Aksa Fathima**, Ujjwal Chaudhary, Ananya Pal, Pavithran. R and Hardik J. Pandya, "**Dye-Integrated Electrospun Nanofibre Sensor for Ammonia Detection: A portable and Low-cost Approach using PCL-BPG/BCG Composite."** XXII International Workshop on the Physics of Semiconductor Devices (IWPSD-2025) IIT-Roorkee, Uttarakhand, India. [Poster Presentation]

RESEARCH OUTCOMES: PATENTS

[1] Invention Title: **A Multiplexed Electrochemical Sensor**
Inventors: Hardik J. Pandya, Anil Vishnu, Hari R. S. Ajay Krishnan, **Aksa Fathima** and Srividyasakti V.

Date of Filing: 13th December 2023 Date of Acceptance: 21st October 2025
Indian patent number: 572340
Current Status: **Granted** Patent Type: **Utility Patent**.

[2] Invention Title: **Asymmetric Interdigitated Electrode Arrays for Enhanced Capacitive Biomolecule Sensing**.
Inventors: Hardik. J Pandya, Aksa Fathima and Hari. R S
Current Status: Application preparation Patent Type: Design & Utility

WORK AND RESEARCH EXPERIENCE

- **Project Associate II in Biomedical Research Lab**

Sept 2022 - Present

Biomedical and Electronic (10^{-6}-10^{-9}) Engineering Systems Laboratory (BEES Lab),
Department of Electronic Systems Engineering, IISc
Advisor: **Prof. Hardik J. Pandya**
Contribution: I am working on the design, development, and validation of solutions to unmet Point-of-care diagnostic problems for congestive heart failure, VOC detection, oral cancer, and oral cancer metastasis using different modalities.

- **Internship in Medical College** *Dec 2021 – Feb 2022*

Centre of Excellence in Molecular Biology and Regenerative Medicine (CEMR) lab,
JSS Medical College, Mysore.
Advisor: **Dr SubbaRao V. Madhunapantula**
Contribution: I worked on Cervical cancer cell line studies to assess the efficacy of the drug research project, where I gained knowledge and assisted with various molecular biology techniques and cell line handling.

TEACHING EXPERIENCE

- Coursera: Sensors and Systems for Biomedical Application
  Topic: Glucose biosensor
- TalentSprit: Sensors Technology [PG level advanced certification programme]
  Topic: Lab session on Thermal and e-beam evaporation

Technical Skills

- **Surface Modification:** Self-assembled monolayer, biofunctionalisation techniques, bioconjugation of biomolecules and nanoparticles.
- **Nanoparticle Synthesis Methods:** Sol-gel method, chemical reduction method, green synthesis methods.
- Molecular-biology Techniques: ELISA, electrophoresis, nucleic acid extraction, and Immunohistochemistry
- **Micro/Nano-Fabrication Techniques (Independent user): Trained user in the cleanroom of class100/1000 (ISO6)**
- Thin Film deposition techniques (Thermal evaporation and e-beam evaporation), Reactive Ion Etching (Cl/F), and Optical Lithography.
- **Electrical Characterization Techniques:** Electrical Impedance Spectroscopy (EIS), Cyclic Voltammetry (CV), Differential Pulse Voltammetry, Square wave Voltammetry (SWV), I-V measurements using 2-probe/4-probe station measurement.
- **Materials Characterization (Independent user):** UV-Visible spectroscopy, FTIR, X-ray spectroscopy (XPS), Dynamic light scattering spectroscopy, Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM).
- **Biomedical Data Analysis**: Data analysis and interpretation using OriginPro, Graphpad Prism and the basics of Python
- **Computational Chemistry-Molecular Modelling and Density Functional Theory (DFT):** Quantum Espresso, NAMD and VESTA software (Developing Proficiency)
- **Additional Skillset:** Technical and Scientific Writing and Presentation Skills, Good Communication Skills, Teamwork

Other Roles and Responsibilities

- **U&I Non-Profitable Organisation** *2019 - 2022*
  Role: Child Protection Policy (CPP) Ambassador to the city chapter
  Prevention of Sexual Harassment (POSH) Ambassador to the city chapter
  Team leader to one of the centres.
- **Health Camp Organizer – All India Institute of Speech and Hearing (AIISH)** *Aug-Sep 2021*
  Role: Volunteer for underprivileged kids with congenital cardiac defects for their routine diagnosis.

References

[1] Prof. Hardik J. Pandya
Associate Professor, Department of Electronics Systems Engineering, Division of EECS,
Principal Investigator, Biomedical and Electronic (10^{-6}-10^{-9}) Engineering Systems Laboratory,
Indian Institute of Science, Bangalore 560012.
Email: hjpandya@iisc.ac.in

[2] Dr. Asha Srinivasan
Assistant Professor, Division of Nanoscience and Technology,
JSS Academy of Higher Education & Research, Mysore 570015
Email: asha.srinivasan@jssuni.edu.in

[3] Dr. Amritha Suresh
Operational Head, Mazumdar Shaw Medical Foundation, Principal Investigator,
Integrated Head and Neck Oncology Program, Mazumdar Shaw Centre for Translational Research,
Narayana Health, Bommasandra Indl Area, Bangalore 560099.
Email: amritha.suresh@gmail.com